Resequencing Considerations in Parallel Downloads

Yoav Nebat and Moshe Sidi
Electrical Engineering Department
Technion, Haifa 32000, Israel
nyoav@tx.technion.ac.il moshe@ee.technion.ac.il

Abstract

Several recent studies have proposed methods to accelerate the receipt of a resource by downloading its parts from different servers in parallel. This paper formulates models for an approach based on receiving only one copy of each of the data packets in a file, while different packets may be obtained from different sources. This approach guarantees faster downloads with lower network use. However, out-of-order arrivals at the receiving side are unavoidable. We present methods to keep out-of-order low to insure more regulated flow of packets to the application. Recent papers indicate that out-of-order arrivals have many unfavorable consequences. A good indicator to the severeness of out-of-order arrival is the resequencing-buffer occupancy. The paper focuses on the analysis of the resequencing-buffer occupancy distribution and on the analysis of the methods used to reduce the occupancy of the buffer.

Keywords

Parallel service; Resequencing-buffer; Mirror sites.

I. INTRODUCTION

TEXT files, pictures, web pages and similar resources are often available at several sites in the network. A user that needs such a resource, say a file, would like to take advantage of the existence of copies of the same file in a manner that will result in better performance (e.g., smaller download time).

The common approach to obtain a resource in today's networks is to open a single TCP connection with a single source where the resource is available and receive all the required data via this single connection. While being very simple, this approach results in slow downloads, since only a single source is used, unless the client's link is utilized to its capacity by this source. With this approach, the fact that the same data is available at more than one location can be exploited by choosing a source that may give advantageous download times - a task that is often complicated. Studies [1,2,3,4] propose techniques for gathering information relevant to server selection.

A more recent approach is to open multiple TCP connections to the various locations where the same data is available, and download the data redundantly in parallel from all the sources. This may be done by applying some coding, e.g.,"Tornado Codes" or "Erasure Codes" [5,6,7,8]. Transmitting a file repeatedly to multicast groups that may be joined by any client that may need the file [5,8,9] takes advantage of the redundancy in sources and seems to be very efficient in the multi-destination case (many-to-many). In the single destination case (many-to-one), it may result in some overhead in the total amount of transmitted data. This in turn may result in greater delay due to congestion and higher paid prices when charges are for the total

amount of data received. Moreover, the flow of data to the application will be rather bursty, due to extensive resequencing that may be needed. In fact, for the codes suggested in [6,7,8] the decoding may begin only after the data is received completely from all sources, resulting in additional delay and highly bursty arrivals to the application.

Another approach is based on the fact that a user should strive to minimize the total download time, while keeping the overall data transmitted as low as possible. This can be achieved by downloading different parts of a file from different sources. Note that with this approach data segments are likely to arrive out-of-order, which may cause large resequencing-buffer demands, and re-ordering, resulting in an unregulated flow of data to the application at the destination [10]. When unicast TCP is used, the occurring of reordering may result in unfavorable consequences, including self-reinforcingly poor performance [11]. An example for this approach is described in [12], where performance was measured for cases where different parts of a file were requested simultaneously and downloaded in parallel from several servers. The results indicate improved performance, compared to a single source case, even if the "best" single source was to be chosen. Analysis of this approach is not available.

This paper focuses on the approach according to which each packet of a file is received from a single source, while different packets may be obtained from different sources. The basic idea is to obtain, before or during the reception process, statistics on the current rate of transmission for all the relevant sources and on the delays on the particular path that each source will use for transmission to the user. This can be achieved by using end-to-end measurements. Based on that information, specific packets are demanded from each source for fast reception of the entire file and for low reception time variance. This approach results in faster reception, lower network use cost, smaller resequencing-buffers, and a more regulated flow of data to the application. The goals of this paper are to formulate suitable models for an approach similar to the latter approach described above and analyze these models in terms of the resequencing-buffer sizes for different file partitions.

In our model, every packet in a downloaded file is assigned to (i.e., requested from) a *single* source responsible for its transmission. Packets may be received out of sequence, even if the packets received from any single source are received in the order transmitted. In this paper, we derive the probability distribution of the resequencing-buffer occupancy for several possible scenarios. This is important for the determination of necessary buffer sizes at the receiver to keep potential overflows under a specific threshold. We also devise algorithms for assigning which packets to download from each source, to help in keeping the resequencing-buffer occupancy as low as possible, and the data flow to the application as smooth as possible. Performance analysis of these algorithms is also presented.

The rest of this paper in organized as follows: In Section II we present the sources, destination and delay

models. In particular, we derive a packet delay model for a single source that guarantees that packets it sends arrive in order to the destination. Section III contains the analysis for homogeneous sources that have the same transmission rates, while Section IV contains the analysis for heterogeneous sources. Section V introduces algorithms for reducing the resequencing-buffer occupancy. Section VI contains some numerical examples. Finally, section VII contains a summary and future work.

II. The model

A. The sources and the destination

We consider the case where N sources transmit different segments of the same file to a destination. Let $\mathcal{L}$ denote the file to be downloaded and $\mathcal{L}_i$ $(1 \leq i \leq N)$ be the segment of the file to be downloaded from source i. To insure that each segment of the file is downloaded from a single source we need that $\mathcal{L}_i \cap \mathcal{L}_j = \emptyset$ for all $i \neq j$. Clearly, $\cup_i \mathcal{L}_i = \mathcal{L}$. An example for a file partition is for $\mathcal{L}_i$ to contain packets $i + lN$ for $l = 0, 1, 2, \ldots$.

One of the main problems dealing with the situation of receiving data segments of the same file from different sources simultaneously is that in order to forward the data to the application at the destination in sequence, the receiver at the destination must store packets received out of sequence in a *resequencing-buffer* until all their preceding packets are received. When data is received from several sources, packets may be received out of sequence, even if packets received from any single source are received in the same order they were sent.

Due to the random delay, the occupancy of the resequencing-buffer at the destination will be random. Our goal is to determine the occupancy probabilities of this buffer. Having the probability distribution of the buffer occupancy will allow to provision the necessary buffer size at the receiver to keep potential overflows under a specified threshold.

B. The delay from source to destination

Let us consider the random delay packets are experiencing when transmitted by a source to the destination. Here we focus on delay models that ensure packets from a single source arrive at their destination in the same order they were transmitted despite their random delay. This is often the case when there is a single static route from the source to the destination. Other delay models, including an analysis of the resequencing-buffer occupancy while receiving data from a single source, are studied in [13].

In the following we derive the basic necessary conditions upon the delay distribution to ensure packets arrive at their destination in the same order they were transmitted. To facilitate the explanation, we use the following notations:

T_i - the time packet i was transmitted

d_i - the delay experienced by packet i

R_i - the time packet i arrived at the destination

Δ_i^t - the inter-departure time for packets $i-1$ and i

Δ_i^r - the inter-arrival time for packets $i-1$ and i

Obviously, $R_i = T_i + d_i$, $\Delta_i^t = T_i - T_{i-1}$ and $\Delta_i^r = R_i - R_{i-1}$. To ensure arrival in sequence it will suffice to ensure that for all $i \geq 0$, packet $i+1$ will not arrive at the destinations' receiver before packet i, since if packet $i+1$ does not arrive before packet i, and packet $i+2$ does not arrive before packet $i+1$, surely packet $i+2$ does not arrive before packet i. Consequently, by induction, no packet transmitted after packet i arrives before packet i, and since this holds for all possible i, the receipt order is the same as the transmission order.

The time packet i is received can be written as $R_i = T_{i-1} + \Delta_i^t + d_i$, while the time packet $i-1$ is received is $R_{i-1} = T_{i-1} + d_{i-1}$. To ensure orderly reception, it suffices to ensure that $\Delta_i^r = R_i - R_{i-1} \geq 0 \; \forall i$. This condition can be written as $\Delta_i^r = R_i - R_{i-1} = T_{i-1} + \Delta_i^t + d_i - T_{i-1} - d_{i-1} = \Delta_i^t + d_i - d_{i-1} \geq 0 \; \forall i$, or $d_i \geq d_{i-1} - \Delta_i^t$. We also need d_i to be non-negative, $d_i \geq 0 \; \forall i$.

Consequently, the only restrictions upon d_i are: $d_i \geq 0$ and $d_i > d_{i-1} - \Delta_i^t$. Therefore, d_i can be any probabilistic function $f(d_{i-1}, \Delta_i^t)$ that satisfies

$$f(d_{i-1}, \Delta_i^t) \geq \begin{cases} d_{i-1} - \Delta_i^t & d_{i-1} - \Delta_i^t \geq 0 \\ 0 & d_{i-1} - \Delta_i^t < 0 \end{cases} \tag{1}$$

To ensure the delay does not diverge, we need a stability constraint on $f(d_{i-1}, \Delta_i^t)$.

For the special case where packets are transmitted in a constant rate, i.e., every time unit, we have $\Delta_i^t = 1 \; \forall i \geq 1$, and (1) becomes

$$f(d_{i-1}, \Delta_i^t) = f(d_{i-1}) \geq \begin{cases} d_{i-1} - 1 & d_{i-1} \geq 1 \\ 0 & d_{i-1} < 1 \end{cases}$$

For simplicity, let the delay be integer multiples of a time unit. Consequently, $f(d_{i-1})$ can be any integer that satisfies

$$d_i = f(d_{i-1}) \geq \begin{cases} d_{i-1} - 1 & d_{i-1} > 0 \\ 0 & d_{i-1} = 0 \end{cases} \tag{2}$$

Any probabilistic function that corresponds to f in (2) can be represented by a Markov-chain, similar to the one in Fig. 1, where state i corresponds to delay of i time units and the arrows correspond to the transitions among states with the respective probabilities. For stability of the delay we need (see Theorem 4 in [14]) that there exists M, such that

$$p_{i(i-1)} \geq \sum_{k=i+1}^{\infty} (k-i) p_{ik} \quad \forall i \geq M$$

which is sufficient to ensure the irreducible and aperiodic Markov-chain that represents the delay process is recurrent.

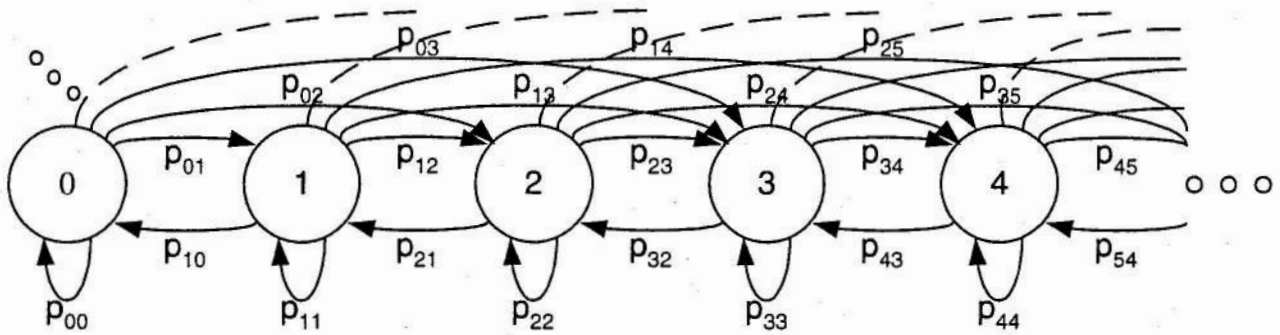


Fig. 1. The Markov-chain representing the delay

III. Homogeneous Sources

In this section we consider homogeneous sources, i.e., sources with the same transmission rate - one packet per time unit. We begin by considering a simple partition of a file in which the packets transmitted at time 0 by sources $1, 2, \ldots, N$ are packets $1, 2, \ldots, N$, respectively. After transmitting the first N packets, the packets are appointed to the sources cyclically, thus, at time t sources $1, 2, \ldots, N$ transmit packets $1 + Nt, 2 + Nt, \ldots, N + Nt$, respectively, i.e., $\mathcal{L}_i$ contains packets $i + lN$ for $l = 0, 1, 2, \ldots$.

The delay of packets from each source is a random process that corresponds to the general discrete delay model for packets received in sequence from a single source. As we showed in Section II-B, the delay process can be represented by a Markov-chain (see Fig. 1). The transition probabilities in the delay model for packets transmitted by source i may be different from those in the delay model for packets transmitted by source j ($i \neq j$), thus, the average delay and the steady-state probabilities may be different. The delays of different sources are independent. Note that we allow zero delays in our model.

A. Two sources

To facilitate the presentation we begin with a two-source system. In this case source 1 transmits the odd indexed packets (i.e., 1,3,5...) while source 2 transmits the even indexed packets (i.e., 2,4,6...). To find the distribution of the resequencing-buffer occupancy we assume steady-state is maintained before time t, and calculate the resequencing-buffer occupancy probabilities at time t. The following definitions are required:

Definition: d_X is the delay experienced by the last packet received from source X.

Definition: Δ_X is the time that passed since receipt of the last packet from source X.

Definition: δ_X is the time that passed since the last packet received from source X was transmitted.

The value d_X may change upon receipt of a packet from source X, and its value is set to the value of the delay of the packet that caused the change. If more than one packet transmitted by source X arrive simultaneously, d_X obtains the value of the delay of the packet most recently sent. The value Δ_X may change

every time unit. If a packet from source X has arrived $\Delta_X = 0$, otherwise Δ_X is increased by 1.

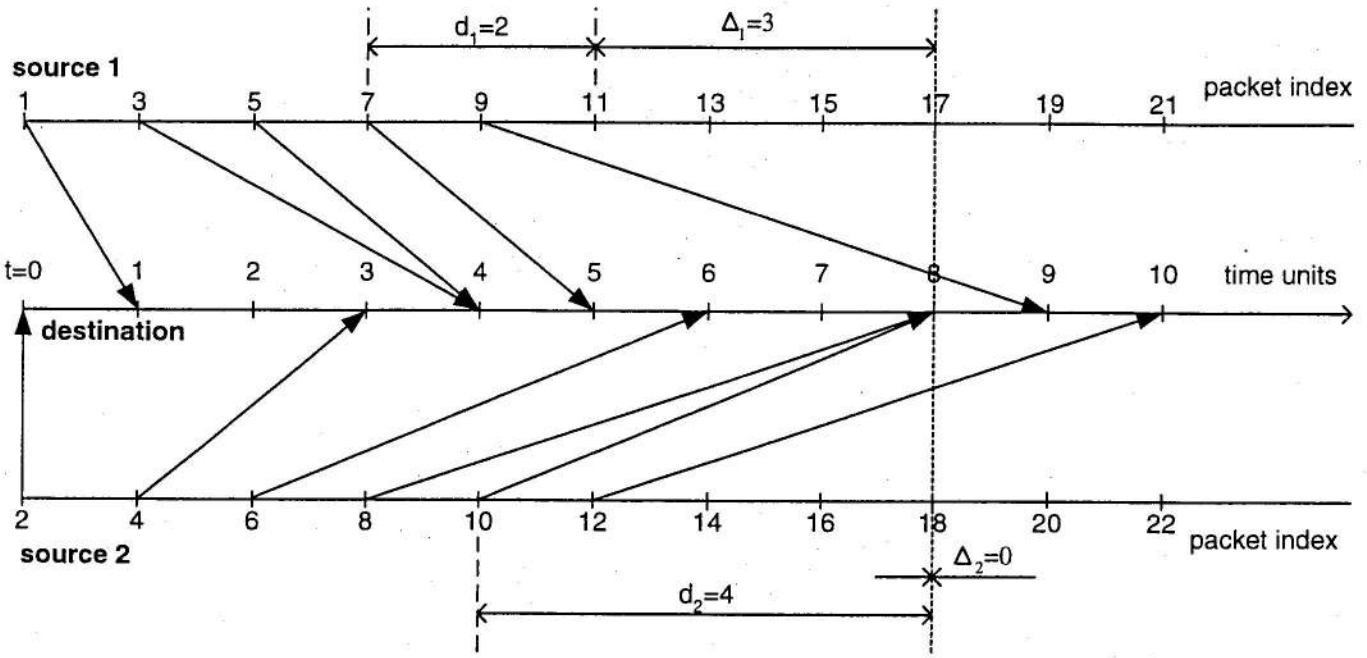


Fig. 2. d_1, d_2, Δ_1 and Δ_2 in a typical scenario

Fig. 2 depicts a typical scenario for two sources. We can see that at time $t = 8$ the last packet received from source 1 is packet 7. This packet was transmitted at time $t = 3$ and arrived at the destination at time $t = 5$, therefore, at time $t = 8$ we have $d_1 = 2$. No packet from source 1 was received between $t = 5$ and $t = 8$, therefore, at time $t = 8$, $\Delta_1 = 3$. On the other hand, the last packets received from source 2 at time $t = 8$ are packets 8 and 10. The most recently transmitted packet among these packets (packet 10) was transmitted at $t = 4$ and arrived at time $t = 8$, therefore, at time $t = 8$ we have: $d_2 = 4$ and $\Delta_2 = 0$. From Fig. 2 and the definition of δ_X it is easy to conclude that $\delta_X = d_X + \Delta_X$.

Theorem 3.1: Given δ_1 and δ_2 at time t, the resequencing-buffer occupancy at time t, B, is given by:

$$B = \begin{cases} \delta_1 - \delta_2 & \delta_1 \geq \delta_2 \\ \delta_2 - \delta_1 - 1 & \delta_1 < \delta_2 \end{cases} \tag{3}$$

Proof: At time t source 1 transmits packet $1 + 2t$, and source 2 transmits packet $2 + 2t$. Given δ_1 and δ_2, the last packet received from source 1 is packet $L_1 = 1 + 2t - 2\delta_1$, and the last packet received from source 2 is packet $L_2 = 2 + 2t - 2\delta_2$.

If $\delta_1 \geq \delta_2$ then $L_2 > L_1$. Therefore, if there are any packets in the resequencing-buffer they are even indexed packets transmitted by source 2, and the resequencing-buffer occupancy is

$$B = \frac{L_2 - (L_1 + 1)}{2} = \delta_1 - \delta_2 \tag{4}$$

since packet $L_1 + 1$ (transmitted by source 2) is not stored in the resequencing-buffer, because all its preceding packets have arrived. All the packets that were transmitted by source 2 after packet $L_1 + 1$, and arrived till

time t are stored in the resequencing-buffer. Since no other packets transmitted by source 2 and no packet transmitted by source 1 are stored in the resequencing-buffer, (4) holds.

If $\delta_1 < \delta_2$ then $L_1 > L_2$, therefore, if there are any packets in the Resequencing-Buffer they are odd indexed packets transmitted by source 1, and the Resequencing-Buffer occupancy is

$$B = \frac{L_1 - (L_2 + 1)}{2} = \delta_2 - \delta_1 - 1 \tag{5}$$

since packet $L_2 + 1$ (transmitted by source 1) is not stored in the Resequencing-Buffer, because all its preceding packets have arrived. All the packets that were transmitted by source 1 after packet $L_2 + 1$, and arrived till time t are stored in the Resequencing-Buffer. Since no other packets transmitted by source 1 and no packet transmitted by source 2 are stored in the Resequencing-Buffer, (5) holds. Therefore, (3) holds, and the proof is completed. □

From (3) we conclude that the steady-state probability of having k packets ($k = 0, 1, 2, \ldots$) stored in the resequencing-buffer is

$$P(B = k) = P(\delta_1 - \delta_2 = k) + P(\delta_2 - \delta_1 - 1 = k) \tag{6}$$

Since δ_1 depends solely on the delay of packets transmitted by source 1, and, δ_2 depends solely on the delay of packets transmitted by source 2, δ_1 and δ_2 are independent. Remembering that $\delta_1 \geq 0$ and $\delta_2 \geq 0$, (6) becomes

$$P(B = k) = \sum_{x=0}^{\infty} P(\delta_2 = x)P(\delta_1 = x + k) + \sum_{x=0}^{\infty} P(\delta_1 = x)P(\delta_2 = x + 1 + k) \tag{7}$$

If we assume the delay of packets transmitted by sources 1 and 2 is limited to M_1 and M_2 respectively, (7) becomes

$$P(B = k) = \sum_{x=0}^{M_2} P(\delta_2 = x)P(\delta_1 = x + k) + \sum_{x=0}^{M_1} P(\delta_1 = x)P(\delta_2 = x + 1 + k) \tag{8}$$

since for any source X, δ_X has an upper limit equal to the maximal delay of packets transmitted by source X.

To find the steady-state probabilities $P(\delta_1 = x)$ and $P(\delta_2 = x)$,$x = 0, 1, 2, \ldots$ we shall prove the following:

Theorem 3.2: The steady-state probability distribution of δ_X is equal to the probability distribution of the delay of packets transmitted by source X.

Proof: In order to prove the above theorem we shall prove that $P(\delta_X = n) = P(D_X = n)$, $n = 0, 1, 2, \ldots$, where D_X is the delay of a packet transmitted by source X.

Definition: A **cycle** is a time period that begins upon transmission of a packet for which $D_X \neq 0$, just after transmission of a packet for which $D_X = 0$, and ends after the transmission of a packet for which $D_X = 0$, where for the next packet to be transmitted $D_X \neq 0$.

A typical cycle is illustrated in Fig. 3.

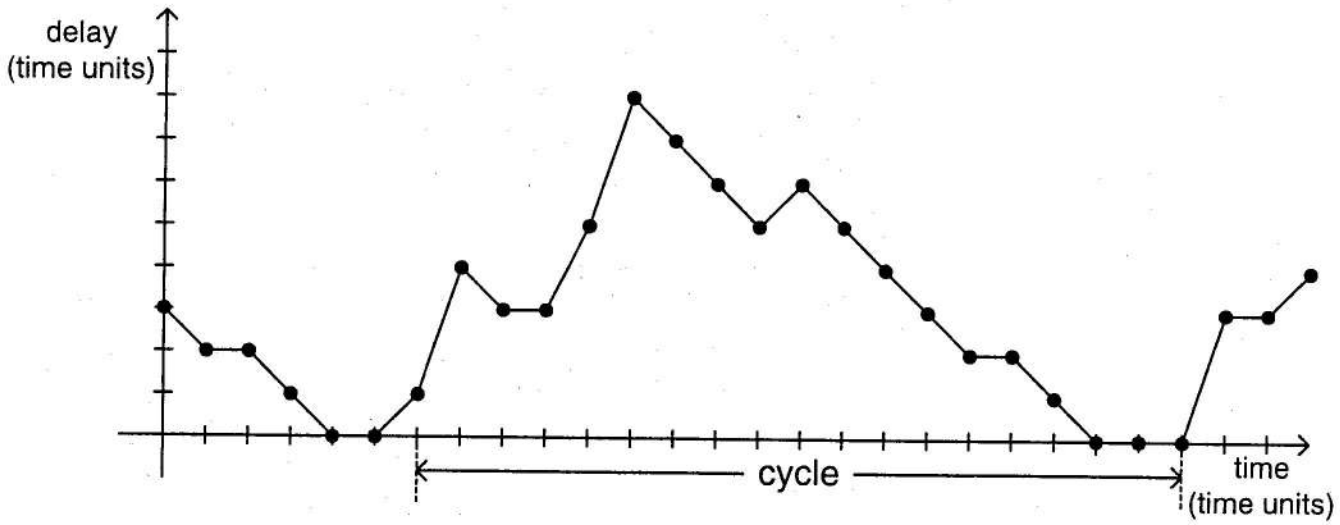


Fig. 3. A typical delay sequence and a cycle

Since the Markov chain that correspond to the delay in all models we consider is stable, the delay process is simply a sequence of cycles. Therefore, in order to prove that $P(\delta_X = n) = P(D_X = n)$, $n = 0, 1, 2, \ldots$ it will suffice to show that the number of times $\delta_X = n$, $n = 0, 1, 2, \ldots$ in every cycle is equal to the number of times $D_X = n$ in the same cycle.

We denote the number of times a packet with delay j is transmitted in a specific cycle, followed by a packet for which the delay is not lower than j by n_1^j. We denote the number of times a packet with delay j is transmitted in a specific cycle, followed by a packet for which the delay is $j-1$ by $n_{\bar{1}}^j$. The total number of times packets for which the delay is j are transmitted is $n^j = n_1^j + n_{\bar{1}}^j$.

Lemma 3.1: If a packet that was transmitted at time t was delayed j time units, and the packet transmitted at time $t+1$ was delayed $j+k$, $k \geq 0$ time units then at time $t+j$ we get $\delta_X = j$, and times t and $t+j$ are in the same cycle.

Proof: The packet transmitted at time t will arrive at the destination at time $t+j$, thus at time $t+j$ we have $\Delta_X = 0$. The packet transmitted at time $t+1$ will arrive at the destination at time $t+1+j+k$, which is after time $t+j$, therefore, no packet transmitted after time t will arrive at the destination till time $t+j$, thus, at time $t+j$ we have $d_X = j$. From the above we conclude that at time $t+j$, $\delta_X \equiv d_X + \Delta_X = j$. A cycle ends after transmission of a packet with delay 0, such a packet could be transmitted not less than j time units after time t, since the delay decreases at most by 1 time unit per time unit. Therefore, time $t+j$ is part of the same cycle. □

We denote the number of times in a specific cycle that a packet with delay lower than j is transmitted, followed by a packet for which the delay is not lower than j by n_2^j.

Lemma 3.2: If a packet that was transmitted at time t was delayed $j-k$, $k>0$ time units, and the packet transmitted at time $t+1$ was delayed $j+i$, $i \geq 0$ time units then at time $t+j$ we get $\delta_X = j$, and times t and $t+j$ are in the same cycle.

Proof: The packet transmitted at time t will arrive at the destination at time $t+j-k$. The packet transmitted at time $t+1$ will arrive at the destination at time $t+1+j+i$. Due to the fact that packets arrive at the destination in the same order they were sent, no packet transmitted by source X could arrive between time $t+j-k$ and time $t+1+j+i$. Therefore, at time $t+j$, $d_X = j-k$ and $\Delta_X = t+j-(t+j-k) = k$. From the above we conclude that at time $t+j$, $\delta_X \equiv d_X + \Delta_X = j-k+k = j$. Given that at time $t+1$ a packet with delay $j+i$ was transmitted, the cycle couldn't end before time $t+1+j+i$, therefore, time $t+j$ is in the same cycle. □

Noticing that since n_1^j is exactly the number of times the delay drops under j time units during the cycle, and n_2^j is exactly the number of times the delay grows to or above j time units during the cycle, we conclude that $n_1^j = n_2^j$, therefore, the total number of times a packet of which the delay is j time units is transmitted during the cycle is $n^j = n_1^j + n_{\bar{1}}^j = n_1^j + n_2^j$. If we denote the number of times that $\delta_X = j$ during the cycle by m^j, it follows from lemma 3.1 and lemma 3.2 that for all $j \geq 0$

$$m^j \geq n_1^j + n_2^j = n^j \tag{9}$$

Now, if we denote the length of the cycle by l, using (9) we have

$$l = \sum_{j=0}^{\infty} m^j \geq \sum_{j=0}^{\infty} n^j = l \tag{10}$$

From (9) and (10) we conclude that $m^j = n^j$, $\forall j \geq 0$. Therefore, the equality

$$P(\delta_X = k) = P(D_X = k)\ ,\ k \geq 0 \tag{11}$$

holds, and the proof of Theorem 3.2 is completed. □

Fig. 4 demonstrates δ_X and D_X in a typical cycle. It can be verified that the number of times the delay is $j, \forall j$ in the cycle is equal to the number of times δ_X equals j in the cycle.

Let P_1^k and P_2^k be the steady-state probabilities of the delay of a packet transmitted by sources 1 and 2, respectively, being k , $k \geq 0$. Substituting (11) into (7) we have

$$P(B=k) = \sum_{x=0}^{\infty} P_2^x P_1^{x+k} + \sum_{x=0}^{\infty} P_1^x P_2^{x+1+k} \tag{12}$$

Assuming the models for the delay of packets transmitted by sources 1 and 2 are identical, and can be represented by the birth-death Markov-chain, shown in Fig. 5, the steady-state probabilities of the delay of packets are

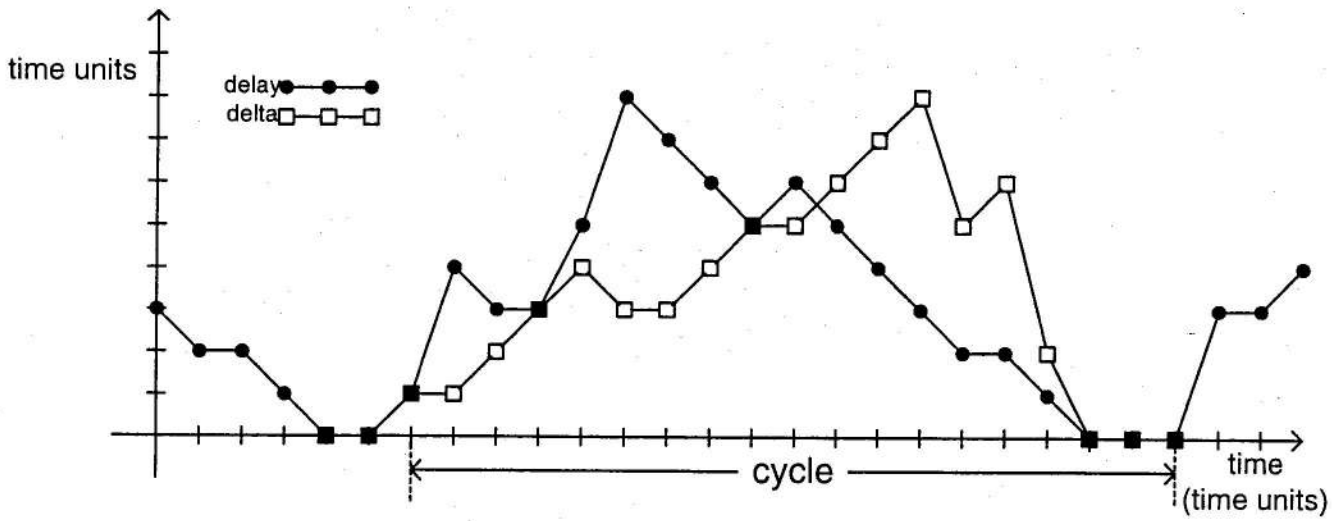


Fig. 4. A typical delay sequence and the corresponding δ_X in a cycle

$$P_1^k = P_2^k = \begin{cases} \alpha\rho^k \frac{p_1}{p_+} & k \geq 1 \\ \alpha & k = 0 \end{cases}$$

where $\rho \triangleq \frac{p_+}{p_-}$, $\rho < 1$ and $\alpha \triangleq \frac{p_- - p_+}{p_- - p_+ + p_1}$.

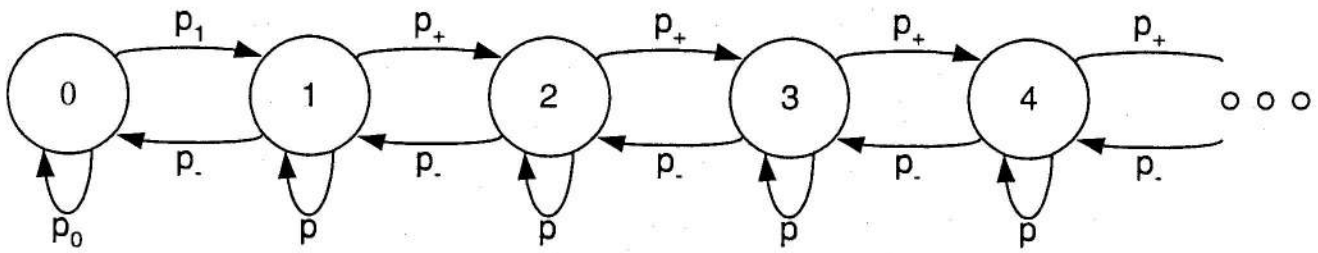


Fig. 5. The simple Markov-chain representing the delay

Substituting P_1^k , $k \geq 0$ and P_2^k , $k \geq 0$ in (12) we obtain

$$P(B = k) = \begin{cases} (\alpha^2\rho^k \frac{p_1}{p_+}(\rho+1)\beta & k \geq 1 \\ (\alpha^2[1 + \frac{p_1}{p_+}((\rho+1)\beta - 1)] & k = 0 \end{cases}$$

where $\beta = 1 + \frac{p_1}{p_+}\frac{\rho^2}{1-\rho^2}$.

B. N sources

We consider the case where N sources transmit different pieces of the same file to a destination. All sources transmit packets in the same rate, one packet per time unit. The packets transmitted at time 0 by sources $1, 2, \ldots, N$ are packets $1, 2, \ldots, N$ respectively. After transmitting the first N packets, the packets are appointed to the sources cyclically, thus, at time t sources $1, 2, \ldots, N$ transmit packets $1+Nt, 2+Nt, \ldots, N+$

Nt respectively, i.e., $\mathcal{L}_i$ contains packets $i + lN$ for $l = 0, 1, 2, \ldots$.

Packets sent by each source are delayed on their route to the destination. The delay of packets from each source is a random process that corresponds to the general discrete delay model for packets received in sequence from a single source. As we mentioned before, A typical Markov-chain representing the delay of packets is represented in Fig. 1.

The derivation of the resequencing-buffer occupancy for N sources is similar to the two sources case. Yet, the notion of the *mvp (minimum valued packet)* is helpful here. The *mvp* at time t is the lowest indexed packet that has not arrived at the destination's receiver by time t (see [15]). For instance, if packets 1 through 15 arrived at the destination's receiver, but packet 16 did not, packet 16 is the *mvp*.

From the definition of the *mvp*, for the model we consider it is easy to observe that: (i) The resequencing-buffer occupancy at time t is exactly the number of packets indexed higher than the *mvp*, that have reached the destination's receiver by time t; (ii) No packet stored in the resequencing-buffer was transmitted by the source of the *mvp*, since packets transmitted by every source arrive in the same sequence they were transmitted.

Lemma 3.3: If $\delta_i \geq \delta_j \; \forall j \in \{1, 2, \ldots, N\}$ and $i < j \; \forall j \mid \delta_j = \delta_i \; , \; j \neq i$, the source of the *mvp* is source i.

proof: By contradiction, we assume that at time t source i satisfies the above inequalities, i.e. $\delta_i \geq \delta_j \; \forall j \in \{1, 2, \ldots, N\}$ and $i < j \; \forall j \mid \delta_j = \delta_i \; , \; j \neq i$, but source $k \; , \; k \neq i$ is the source of the *mvp* packet.

If we denote the lowest indexed packet transmitted by source x that did not arrive at the destinations receiver by time t by $m^x(t)$, for sources i and k we have: $m^i(t) = i + N(t - \delta_i + 1)$ and $m^k(t) = k + N(t - \delta_k + 1)$, respectively , as shown in Fig. 6.

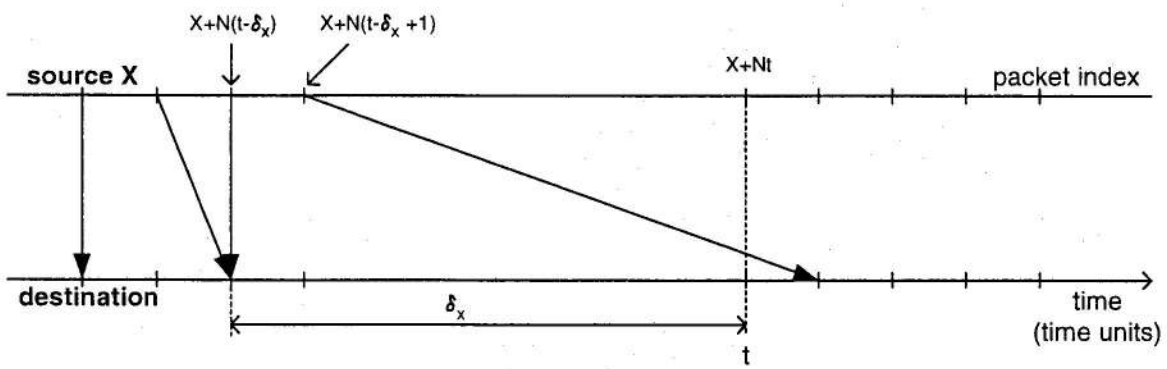


Fig. 6. δ_X and the corresponding lowest indexed packet transmitted by source x that did not arrive at the destinations' receiver

If $k > i$, using the fact that $\delta_i \geq \delta_k$ we get $m^k(t) = k + N(t - \delta_k + 1) \geq k + N(t - \delta_i + 1) > i + N(t - \delta_i + 1) = m^i(t)$, therefore, $m^i(t) < m^k(t)$, contradicting the assumption that the *mvp* packet was a packet transmitted by source k.

On the other hand, if $k < i$, using the fact that $\delta_i > \delta_k$, we get $m^k(t) = k + N(t - \delta_k + 1) = (k + N) + N(t - (\delta_k + 1) + 1) \geq (k + N) + N(t - \delta_i + 1) > i + N(t - \delta_i + 1) = m^i(t)$, therefore, $m^i(t) < m^k(t)$, contradicting

the assumption that the *mvp* packet was a packet transmitted by source k. Therefore, we conclude that no source k, $k \neq i$ could be the source of the *mvp* packet. □

Lemma 3.4: If the source of the *mvp* at time t is i, given δ_i and δ_j, the number of packets that were transmitted by source j, and are stored in the resequencing-buffer, a_j, is given by:

$$a_j = \begin{cases} \delta_i - \delta_j & j > i \\ \delta_i - \delta_j - 1 & j < i \\ 0 & j = i \end{cases} \tag{13}$$

Proof: The index of the *mvp* packet is $m^i(t) = i + N(t - \delta_i + 1)$. The index of the last packet received from source j is $j + N(t - \delta_j)$.

If $j > i$, the lowest indexed packet transmitted by source j and indexed higher than the *mvp* packet is packet $j + N(t - \delta_i + 1)$, therefore, a_j is given by

$$a_j = \frac{j + N(t - \delta_j) - [j + N(t - \delta_i + 1)]}{N} + 1 = \delta_i - \delta_j - 1 + 1 = \delta_i - \delta_j \tag{14}$$

If $j < i$, the lowest indexed packet transmitted by source j and indexed higher than the *mvp* packet is packet $j + N(t - \delta_i + 2)$, therefore, a_j is given by

$$a_j = \frac{j + N(t - \delta_j) - [j + N(t - \delta_i + 2)]}{N} + 1 = \delta_i - \delta_j - 2 + 1 = \delta_i - \delta_j - 1 \tag{15}$$

As mentioned before, no packet stored in the resequencing-buffer was transmitted by the source of the *mvp* packet, therefore, for $j = i$ we have $a_j = 0$. Consequentially, (13) holds, and the proof is completed. □

The resequencing-buffer occupancy is, therefore, given by

$$B = \sum_{j=1}^{i-1} (\delta_i - \delta_j - 1) + \sum_{j=i+1}^{N} (\delta_i - \delta_j) = N\delta_i + 1 - i - \sum_{j=1}^{N} \delta_j \tag{16}$$

Let P_X^k $(k \geq 0)$ denote the steady-state probability of the delay of a packet transmitted by sources X to be k. From Theorem 3.2 we have

$$P_X^k = P(D_X = k) = P(\delta_X = k) = P(d_X + \Delta_X = k) \tag{17}$$

If we denote the source of the *mvp* by s_m, the steady state probability of having k packets stored in the resequencing-buffer is given by

$$P(B = k) = \sum_{i=1}^{N} \sum_{x=0}^{\infty} P(B = k | s_m = i, \delta_i = x) \cdot P(s_m = i, \delta_i = x) \tag{18}$$

The probability $P(s_m = i, \delta_i = x)$ is the probability that $\delta_i = x$, and $\delta_j < \delta_i \; \forall j \in \{1, 2, \ldots, i-1\}$, and $\delta_j \leq \delta_i \; \forall j \in \{i+1, i+2, \ldots, N\}$. The probability that $\delta_i = x$ is simply P_i^x. Defining $F_j(x) \triangleq \sum_{l=0}^{x} P_j^l$ and using the fact that all delay processes are independent we have

$$P(s_m = i, \delta_i = x) = P_i^x \prod_{j=1}^{i-1} F_j(x-1) \prod_{j=i+1}^{N} F_j(x) \tag{19}$$

The steady state probability of having k packets stored in the resequencing-buffer, given that the source of the mvp is i and $\delta_i = x$, using (16) is

$$P(B = k | s_m = i, \delta_i = x) = \tag{20}$$

$$P(Nx + 1 - i - \sum_{j=1}^{N} \delta_j = k | s_m = i, \delta_i = x) =$$

$$P(\sum_{j=1, j\neq i}^{N} \delta_j = (N-1)x + 1 - i - k | s_m = i, \delta_i = x)$$

(20) becomes after some algebra

$$P(B = k | s_m = i, \delta_i = x) = P(\sum_{j=1, j\neq i}^{N} \delta_j = M | i, x, \delta_j < x \; \forall j < i, \delta_j \leq x \; \forall j > i) \tag{21}$$

and (21) can be written as

$$P(B = k | s_m = i, \delta_i = x) = \tag{22}$$

$$\sum_{l_1=0}^{\alpha_1} \sum_{l_2=0}^{\alpha_2} \cdots \sum_{l_{N-1}=0}^{\alpha_{N-1}} \left[\left(\prod_{j=1}^{i-1} \frac{P_j^{l_j}}{F_j(x-1)} \right) \left(\prod_{j=i+1}^{N} \frac{P_j^{l_j}}{F_j(x)} \right) \right]$$

Where α_j is defined by:

$$\alpha_j = \begin{cases} min\{M - \sum_{k=1}^{j-1} l_k, x-1\} & j < i \\ min\{M - \sum_{k=1, k\neq i}^{j-1} l_k, x\} & j > i \end{cases}$$

Where $M = (N-1)x + 1 - i - k$, and for $j = N$ we define $l_N \triangleq M - \sum_{k=1, k\neq i}^{N-1} l_k$.

Finally, the probabilities $P(B = k), (k \geq 0)$ are obtained from (18) by substituting (19) and (22).

IV. Heterogeneous Sources

In this section we focus on a system that consists of two sources with different transmission rates. Source 1 transmits packets with a packet rate of a packets per time unit, while source 2 transmits packets with a packet rate of b packets per time unit. In this case, assigning packets to be transmitted to the sources in the same

manner as in section III-A (i.e., odd indexed packets to source 1, and even indexed packets to source 2), will result in a constant growth of the difference in the indexes of the packets transmitted by the sources, which will cause a steadily growing resequencing-buffer occupancy. Therefore, we should first devise a packet assignment that will result in steady buffer occupancy probabilities, and then find the occupancy probabilities.

In order to simplify the presentation, we assume a and b are integers, with no common divisor greater than 1 (given the rates are rational numbers, we could always define the time unit to be used, so that this will hold).

Packets transmitted by each source are delayed on their route to the destination. The delay of packets from each source is a random process that corresponds the Markov-chain represented in Fig. 1.

A. Packet assignment

The assignment of packets to each source follows the rule that a packet is assigned to the source that would be able to transmit it the earliest. In the case both sources could transmit it at the same time, the packet is assigned to source 1, and the next packet in sequence is assigned to source 2. The algorithm below will assure assignment that corresponds to the conditions mentioned above:

1: define two counters: n, m , and initialize them to be equal to 1.

2: assign packet 1 to source 1, and packet 2 to source 2.

3: If $n > m \cdot \frac{a}{b}$, assign packet $n + m + 1$ to source 2 and increase m by 1. Otherwise, assign packet $n + m + 1$ to source 1 and increase n by 1.

4: If $n + m \leq a + b$ go back to 3.

5: For all $p \in \{1, 2, \ldots, a + b\}$ assign packets $p + k(a + b)$, $k = 1, 2, \ldots$ to the source p was assigned to.

An example of the assignment obtained for $a = 5$ and $b = 3$ is presented in Fig. 7.

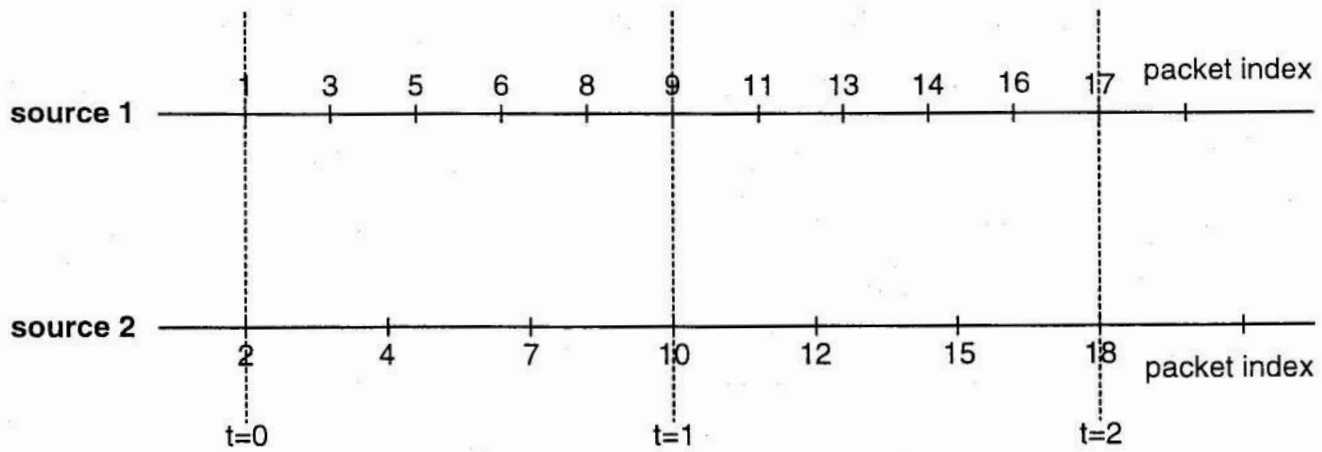


Fig. 7. Packet assignment for $a = 5$ and $b = 3$

B. The resequencing-buffer occupancy

In the Markov-chains representing the delay of packets transmitted by source 1 and source 2, transitions occur upon every packet transmission. Therefore, the transition rates differ, corresponding to the transition rates of the sources. We define a source X time unit as the time between two subsequent packet transmissions of source X. Clearly, a (global) time unit is equal to a source 1 time units, or to b source 2 time units.

In this section we use the following definition for δ_X:

Definition: δ_X is the number of source X time units that passed since the last packet received from source X was transmitted.

Theorem 4.1: The resequencing-buffer occupancy, at the instances that both sources transmit packets simultaneously (every (global) time unit), given δ_1 and δ_2, is given by

$$B = \begin{cases} \lceil \frac{(\delta_2-1)a}{b} \rceil - \delta_1 & a\cdot\delta_2 - b\cdot\delta_1 > a-b \\ \lfloor \frac{(\delta_1-1)b}{a} + 1 \rfloor - \delta_2 & a\cdot\delta_2 - b\cdot\delta_1 \leq a-b \end{cases} \tag{23}$$

Proof: We assume the instance we observe the resequencing-buffer occupancy is t (global) time units.

The last packets that arrived at the destinations' receiver from sources 1 and 2 were transmitted at times $t-\frac{\delta_1}{a}$ and $t-\frac{\delta_2}{b}$, respectively. The *mvp* packet was transmitted at time $min\{t-\frac{\delta_1-1}{a}, t-\frac{\delta_2-1}{b}\}$. Packets stored in the resequencing-buffer are packets that arrived at the destinations' receiver, and were transmitted by the source that did not transmit the *mvp* packet, after the *mvp* packet was transmitted.

The *mvp* packet was transmitted by source 2 if $min\{t-\frac{\delta_1-1}{a}, t-\frac{\delta_2-1}{b}\} = t-\frac{\delta_2-1}{b}$, and $t-\frac{\delta_1-1}{a} \neq t-\frac{\delta_2-1}{b}$. This occurs if and only if $a\delta_2 - b\delta_1 > a-b$. In this case all packets transmitted by source 1 in the time interval $(t-\frac{\delta_2-1}{b}, t-\frac{\delta_1}{a}]$ are stored in the resequencing-buffer. Therefore, for $a\delta_2 - b\delta_1 > a-b$, the number of packets stored in the resequencing-buffer is

$$B = \left\lceil \left[(t-\frac{\delta_1}{a}) - (t-\frac{\delta_2-1}{b}) \right] a \right\rceil = \left\lceil \frac{(\delta_2-1)a}{b} \right\rceil - \delta_1 \tag{24}$$

The *mvp* packet was transmitted by source 1 if $min\{t-\frac{\delta_1-1}{a}, t-\frac{\delta_2-1}{b}\} = t-\frac{\delta_1-1}{a}$, even if $t-\frac{\delta_1-1}{a} = t-\frac{\delta_2-1}{b}$. This occurs if and only if $a\delta_2 - b\delta_1 \leq a-b$. In this case all packets transmitted by source 2 in the time interval $[t-\frac{\delta_1-1}{a}, t-\frac{\delta_2}{b}]$ are stored in the resequencing-buffer. Therefore, for $a\delta_2 - b\delta_1 \leq a-b$, the number of packets stored in the resequencing-buffer is

$$B = \left\lfloor \left[(t-\frac{\delta_2}{b}) - (t-\frac{\delta_1-1}{a}) \right] b + 1 \right\rfloor = \left\lfloor \frac{(\delta_1-1)b}{a} + 1 \right\rfloor - \delta_2 \tag{25}$$

From (24) and (25) we conclude that (23) holds. □

The probability of having k packets stored in the resequencing-buffer is

$$P(B=k) = P(\lceil (\delta_2-1)\frac{a}{b} \rceil - \delta_1 = k\ ,\ a\delta_2 - b\delta_1 > a-b) +$$

$$P(\lfloor(\delta_1-1)\frac{b}{a}+1\rfloor-\delta_2=k\ ,\ a\delta_2-b\delta_1\leq a-b)=$$

$$=\sum_{v=0}^{\infty}P(\lceil(\delta_2-1)\frac{a}{b}\rceil-\delta_1=k\ ,\ a\delta_2-b\delta_1>a-b|\delta_1=v)P(\delta_1=v)+$$

$$\sum_{u=0}^{\infty}P(\lfloor(\delta_1-1)\frac{b}{a}+1\rfloor-\delta_2=k\ ,\ a\delta_2-b\delta_1\leq a-b|\delta_2=u)P(\delta_2=u)=$$

$$=\sum_{v=0}^{\infty}P[k+v-1<(\delta_2-1)\frac{a}{b}\leq k+v\ ,\ \delta_2>1+\frac{b}{a}(v-1)]P(\delta_1=v)+$$

$$\sum_{u=0}^{\infty}P[k+u-1\leq(\delta_1-1)\frac{b}{a}<k+u\ ,\ \delta_1\geq 1+\frac{a}{b}(u-1)]P(\delta_2=u)=$$

$$P(B=k)=\sum_{v=0}^{\infty}P(\frac{(k+v-1)b}{a}+1<\delta_2\leq\frac{(k+v)b}{a}+1)P(\delta_1=v)+ \tag{26}$$

$$\sum_{u=0}^{\infty}P(\frac{(k+u-1)a}{b}+1\leq\delta_1<\frac{(k+u)a}{b}+1)P(\delta_2=u)$$

If we use the definition of P_X^k as in (17) in order to define the PDF of the delay of packets transmitted by source X we have

$$F_X^\omega=P(D_X\leq\omega)=\sum_{j=0}^{\lfloor\omega\rfloor}P_X^j \tag{27}$$

Using Theorem 3.2 and the definition of F_X^ω in (27) we have $P(\delta_X\leq\omega)=F_X^\omega$. Therefore, (26) becomes

$$\begin{aligned}P(B=k)\ &=\ \sum_{v=0}^{\infty}(F_2^{\frac{(k+v)b}{a}+1}-F_2^{\frac{(k+v-1)b}{a}+1})P_1^v\\&+\ \sum_{u=0}^{\infty}(F_1^{\frac{(k+u)a}{b}+1}-\pi_1^{\frac{(k+u)a}{b}+1}-F_1^{\frac{(k+u-1)a}{b}+1}+\pi_1^{\frac{(k+u-1)a}{b}+1})P_2^u\end{aligned}$$

where $\pi_X^\omega=0$ when ω is not an integer, and $\pi_X^\omega=P_X^\omega$ otherwise.

V. Reducing the buffer occupancy

Using the packet assignment method, as in Section III, according to which packets $1, 2, \ldots, N$ are transmitted by sources $1, 2, \ldots, N$, respectively, and afterwards assigning packets cyclically, ignores any knowledge regarding the delay distribution of packets received from every source.

To maintain the resequencing-buffer occupancy as low as possible, we present a new packet assignment method, taking packet delay distribution into consideration. In this section we assume an end-to-end measurement had taken place, and an estimation of the average delay from each source to the destination is available.

A. Packet assignment for two homogeneous sources

To facilitate the presentation we begin with two homogeneous sources, source 1 and source 2, transmitting packets with a rate of one packet per time unit. Let the delay of packets transmitted by each source be constant, i.e., packets transmitted by source 1 are delayed d_1 time units on their route to the destination, and packets transmitted by source 2 are delayed d_2 time units on their route to the destination. Under constant delay conditions, a packet from each source is received every time unit, thus, $\Delta_1 = \Delta_2 = 0$. As a result, $\delta_1 = d_1$ and, $\delta_2 = d_2$, since $\delta_X = d_X + \Delta_X \ \ \forall X$. Consequently, the resequencing-buffer occupancy, using the assignment method represented in Section III-A is, according to (3), given by

$$B = \begin{cases} d_1 - d_2 & d_1 \geq d_2 \\ d_2 - d_1 - 1 & d_1 < d_2 \end{cases} \tag{28}$$

From (28) we conclude that using the packet assignment as in Section III-A results in a constant buffer occupancy, that corresponds approximately to the delay difference, even though the delays are constant.

In the situation represented above, it would be much more advantageous to assign higher indexed packets to the source with the greater delay. For instance, assuming $d_1 > d_2$, and $d_1 - d_2 = \alpha$, if we assign packets $\alpha + 1, \alpha + 3, \alpha + 5, \ldots$ to source 1, and, packets $1, 2, \ldots, \alpha, \alpha + 2, \alpha + 4, \ldots$ to source 2, the resequencing-buffer occupancy would be 0 through out the receipt of the file, since packets 1 through α will be received in order, and then packets $\alpha + 1$ and $\alpha + 2$ will be received simultaneously, as well as packets $\alpha + 3$ and $\alpha + 4$, packets $\alpha + 5$ and $\alpha + 6$, etc.

From the example above we conclude that given the delay distributions of packets from different sources, we can achieve a lower average buffer occupancy if we take the delay distributions into consideration when we assign packets to sources.

The criterion for packet assignment in our algorithm is that packets are assigned to the sources in a manner that ensures that if all the packets experience a delay that corresponds to the average delay of packets transmitted by the source they are assigned to, the buffer occupancy would be 0 throughout the receipt of the file.

In the general case, when packet delay is random, let the average delays of packets transmitted by sources 1 and 2 be $\overline{d}_1$ and $\overline{d}_2$, respectively. We assume, without loss of generality that $\overline{d}_1 \geq \overline{d}_2$, and define $\alpha = \lceil \overline{d}_1 - \overline{d}_2 \rceil$. Similarly to the method shown for constant delays, we assign packets $\alpha+1, \alpha+3, \alpha+5, \ldots$ to source 1, and, packets $1, 2, \ldots, \alpha, \alpha+2, \alpha+4, \ldots$ to source 2.

If we use this packet assignment and assume the delay of packets transmitted by each source is exactly the average delay of the transmitting source, all packets arrivals are in sequence. If packets $\alpha+1$ and 1 are transmitted at time 0 by sources 1 and 2, respectively, and every source transmits a new packet assigned to it every time unit, packets $1, 2, \ldots, \alpha$ would arrive at the destination's receiver at time $\overline{d}_2, \overline{d}_2+1, \ldots, \overline{d}_2+\alpha-1$, respectively. Packet $\alpha+1$ would arrive at the destination's receiver at time $\overline{d}_1, \overline{d}_2+\alpha-1 < \overline{d}_1 \leq \overline{d}_2+\alpha$, and packets $\alpha+3, \alpha+5, \ldots$ would arrive at time $\overline{d}_1+1, \overline{d}_1+2, \ldots$, respectively. Packet $\alpha+2$ would arrive at the destination's receiver at time $\overline{d}_2+\alpha$, and packets $\alpha+4, \alpha+6, \ldots$ would arrive at time $\overline{d}_2+\alpha+1, \overline{d}_2+\alpha+2, \ldots$, respectively. Therefore, packet $\alpha+1$ will be received before packet $\alpha+2$, but after packet α, since $\overline{d}_2+\alpha-1 < \overline{d}_1 \leq \overline{d}_2+\alpha$. Similarly, packet $\alpha+3$ will be received before packet $\alpha+4$, but after packet $\alpha+2$, etc. Consequently, the resequencing-buffer occupancy would be 0 throughout the receipt of the file, since all packets would be received in order, in accordance to the assignment criterion.

Fig. 8 demonstrates how the assignment provides an occupancy of 0, throughout the receipt of all the packets, if all packets are delayed according to the average delay of packets transmitted by their source. In this example $\overline{d}_1 = 4.4$ and $\overline{d}_2 = 2.6$, therefore $\alpha = \lceil \overline{d}_1 - \overline{d}_2 \rceil = 2$. The packets assigned to source 1 are packets $3, 5, 7, \ldots$, and the packets assigned to source 2 are 1 and 2, and then $4, 6, 8, \ldots$. From the figure we can see that packets arrive at the destination in sequence, therefore the resequencing-buffer occupancy is 0 throughout the receipt of the packets.

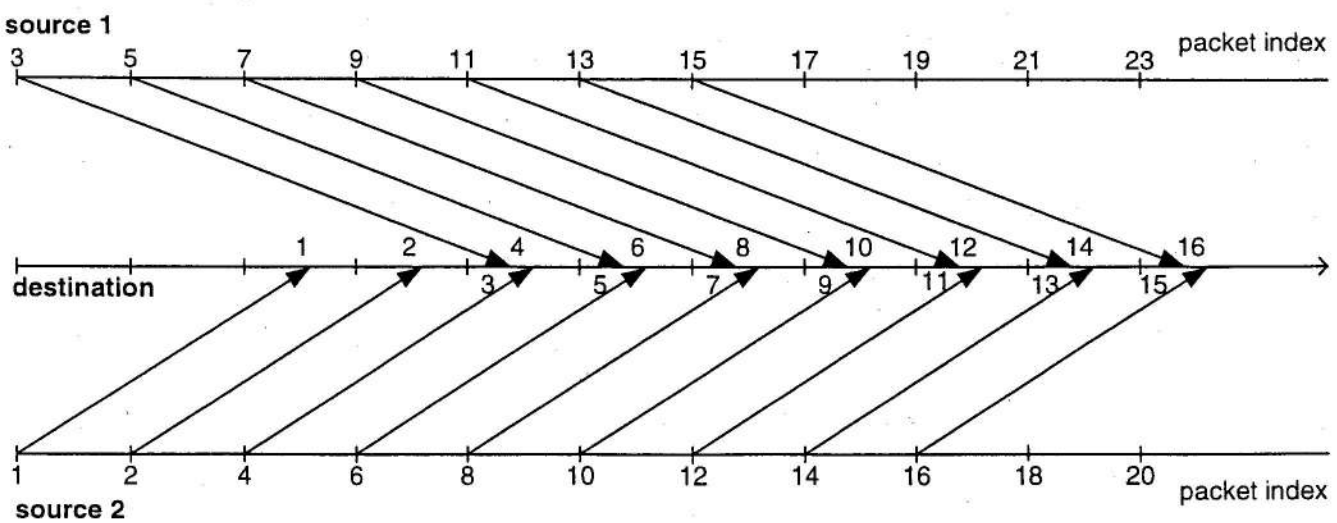


Fig. 8. Packet assignment to sources and arrival times

Lemma 5.1: For the assignment algorithm described above, the resequencing-buffer occupancy probabilities are equal to the resequencing buffer occupancy probabilities, when packets are assigned cyclically, and the delay in all states of the Markov-chain representing the delay of packets transmitted by source 2 is increased by α.

Proof: We shall refer to the two algorithms by: "the packet assignment algorithm", and, "the cyclic assignment-shifted delay algorithm", respectively.

Given δ_1 and δ_2 at time $t > max\{M_1, M_2 + \alpha\}$ (this ensures packets that were assigned in the cyclic assignment part of the packet assignment algorithm to both sources have begun to arrive), the resequencing-buffer occupancy, B, using the packet assignment algorithm is given by:

$$B = \begin{cases} \delta_1 - \delta_2 - \alpha & \delta_1 - \delta_2 \geq \alpha \\ \delta_2 - \delta_1 - 1 + \alpha & \delta_1 - \delta_2 < \alpha \end{cases} \tag{29}$$

The proof is very similar to the proof of theorem 3.1, for details see Appendix.

We denote the steady-state probability that the delay of packets transmitted by source 2 is x, in the cyclic assignment-shifted delay algorithm, by $\tilde{P}_2^x$. We denote the steady-state probability that the delay of packets transmitted by source 2 is x, in the packet assignment algorithm, by P_2^x, and, in both cases, the steady-state probability that the delay of packets transmitted by source 1 is x, by P_1^x.

From the definition of $\tilde{P}_2^x$ and P_2^x we conclude that

$$P_2^x = \tilde{P}_2^{x+\alpha} \quad 0 \leq x \leq M_2 \tag{30}$$

Substituting P_2^x by $\tilde{P}_2^x$ in (12) the probability that the resequencing-buffer occupancy is k, for the cyclic assignment-shifted delay algorithm can be written as

$$\tilde{P}(B = k) = \sum_{x=\alpha}^{M_2+\alpha} \tilde{P}_2^x P_1^{x+k} + \sum_{x=0}^{M_1} P_1^x \tilde{P}_2^{x+1+k} \tag{31}$$

By substituting $\tilde{P}_2^x$ by $P_2^{x-\alpha}$, according to (30), (31) becomes

$$\tilde{P}(B = k) = \sum_{x=\alpha}^{M_2+\alpha} P_2^{x-\alpha} P_1^{x+k} + \sum_{x=0}^{M_1} P_1^x P_2^{x+1+k-\alpha} = \sum_{x=0}^{M_2} P_2^x P_1^{x+k+\alpha} + \sum_{x=0}^{M_1} P_1^x P_2^{x+1+k-\alpha} \tag{32}$$

The probability that the resequencing-buffer occupancy is k, for the packet assignment algorithm, according to (29) is

$$P(B = k) = \sum_{x=0}^{M_2} P(\delta_2 = x) P(\delta_1 = x + k + \alpha) + \sum_{x=0}^{M_1} P(\delta_1 = x) P(\delta_2 = x + 1 + k - \alpha) \tag{33}$$

Applying theorem 3.2, (33) becomes

$$P(B=k)=\sum_{x=0}^{M_2} P_2^x P_1^{x+k+\alpha}+\sum_{x=0}^{M_1} P_1^x P_2^{x+1+k-\alpha} \tag{34}$$

from (32) and (34) we have $P(B=k)=\tilde{P}(B=k)$, and the proof is completed. □

Using Lemma 5.1 we can apply (12) in order to find the resequencing-buffer occupancy probabilities, when packets are assigned to sources according to the packet assignment algorithm.

B. Packet assignment for N homogeneous sources

The assignment algorithm for the case of N homogeneous sources is similar to the algorithm for packet assignment to two sources.

In order to assign packets we define the following algorithm:

Algorithm 1

1: denote the sources, sorted according to their average delay, from lowest to highest by $S_1, S_2, \ldots, S_N$, respectively.

2: denote the average delay of packets transmitted by sources $S_1, S_2, \ldots, S_N$ by $\overline{d}_1, \overline{d}_2, \ldots, \overline{d}_N$, respectively.

3: $\forall i$

$x_i \leftarrow \overline{d}_i$

$L_i \leftarrow 0$

4: $j \leftarrow 0$

5: $j \leftarrow j+1$

6: let S_i be the source with minimum x_i:

assign packet j to source S_i (i.e., $j \in \mathcal{L}_i$)

$x_i \leftarrow x_i+1$

$L_i \leftarrow j$

7: while $L_N = 0$ go to 5

8: for each source i, assign packets $L_i+N, L_i+2N, L_i+3N, \ldots$ to source i (i.e., $\mathcal{L}_i$ contains packets L_i+lN for $l=0,1,2,\ldots$. For $1 \le i \le N-1$, $\mathcal{L}_i$ also contains $\alpha_i - 1$ additional packets, that were assigned to it before packet L_i).

Algorithm 1 is in accordance with the criterion that packets are assigned to the sources in a manner so that if all the packets experience a delay that corresponds to the average delay of packets transmitted by the source they are assigned to, the buffer occupancy would be 0 throughout the receipt of the file. The variables x_i , $1 \le i \le N$ keep the time the next packet assigned to source S_i will be received, if it experiences the average delay of source S_i. Therefore, by assigning each packet to a source S_i if and only if $x_i = \min_{1 \le j \le N} x_j$, we ensure

that if all packets are delayed according to the average delay of the sources they are assigned to, each packet will arrive before all its successors. If we apply this to all packets, by induction, all packets will be received in order, therefore, the resequencing-buffer occupancy will be 0 throughout the receipt of the packets.

As in the packet assignment algorithm, presented in III-B, where after a transient, packets are assigned cyclically to the sources, in Algorithm 1, packets are assigned cyclically, once $L_N \neq 0$. By defining $\alpha_i = \lceil \overline{d}_N - \overline{d}_i \rceil$, $\forall i$, we notice that exactly α_i packets will be assigned to source S_i, before the first packet is assigned to source S_N. Afterwards, packets are assigned to the sources cyclically.

Fig. 9 demonstrates how the assignment provides an occupancy of 0, throughout the receipt of all the packets, for a case where $N = 3$, if all packets are delayed according to the average delay of packets transmitted by their source. In this example the average delays of the sources are 2.6, 4.4 and 5.7. After applying the first step of Algorithm 1, we have that packets transmitted by sources S_1, S_2 and S_3 experience average delays $\overline{d}_1 = 2.6$, $\overline{d}_2 = 4.4$, and $\overline{d}_3 = 5.7$, respectively. Therefore, $\alpha_1 = \lceil \overline{d}_N - \overline{d}_1 \rceil = 4$ and $\alpha_2 = \lceil \overline{d}_N - \overline{d}_2 \rceil = 2$. If we run Algorithm 1, the packets assigned to source S_1 are packets $1, 2, 4, 6$, and then packets $9, 12, 15, \ldots$; the packets assigned to source S_2 are 3 and 5, and then $8, 11, 14, \ldots$; the packets assigned to source S_3 are $7, 10, 13, \ldots$. During the packet assignment, L_3 is changed from 0 to 7, when packet 7 is assigned to source S_3. After packet 7 is assigned, packets are assigned cyclically to the sources. Furthermore, we notice that exactly $\alpha_1 = 4$ packets are assigned to source S_1 before the first packet is assigned to source S_3, and exactly $\alpha_2 = 2$ packets are assigned to source S_2 before the first packet is assigned to source S_3.

From the figure we can verify that packets arrive at the destination in sequence, therefore, the resequencing-buffer occupancy is 0 throughout the receipt of the packets.

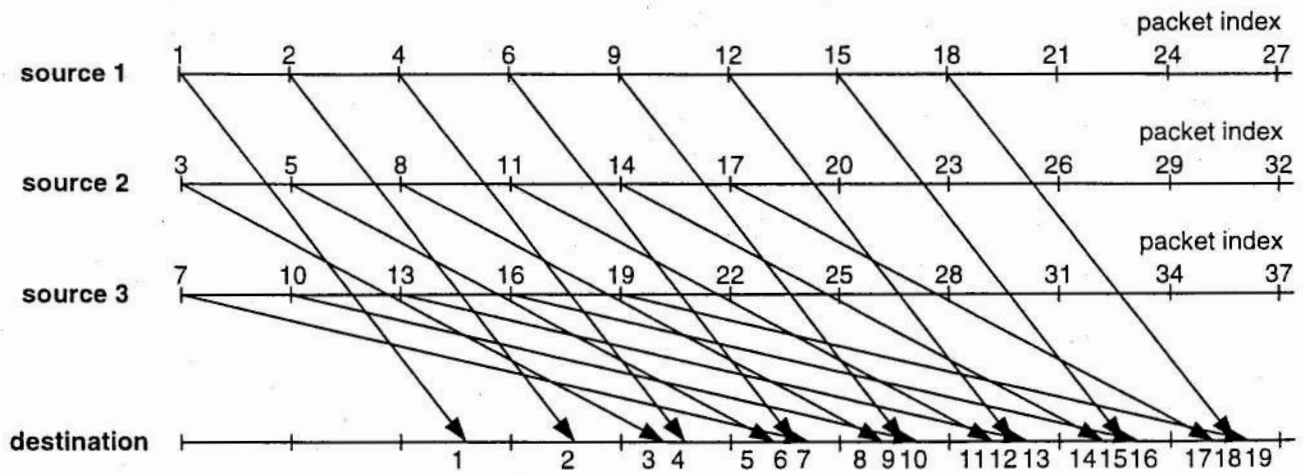


Fig. 9. Packet assignment to 3 sources and arrival times

In order to obtain the resequencing-buffer occupancy probabilities we define an auxiliary algorithm as fol-

lows:

Algorithm 2

1: Increase the delays in all states of the Markov-chain representing the delay of packets transmitted by source $i, 1 \leq i \leq N$ by α_i.

2: Assign packets cyclically. The assignment order is from the source with the minimal resulting average delay to the source with the maximal resulting average delay.

Theorem 5.1: The resequencing-buffer occupancy probabilities, using Algorithm 1 and Algorithm 2 are equal.

The proof of the theorem contains three steps. In the first step (lemma 5.2 below) we establish the linkage between the two algorithms by proving that if packet y is transmitted by source S_i, using Algorithm 2, then packet $y + A$ where $A = \sum_{i=1}^{N} \alpha_i$ is transmitted by source S_i, using Algorithm 1.

In the second step we define the vector $\vec{a} = (a_1, a_2, \ldots, a_N)$ where a_i is the last packet received from S_i, using Algorithm 2. Similarly, $\vec{b} = (A + a_1, A + a_2, \ldots, A + a_N)$ where $A + a_i$ is the last packet received from S_i, using Algorithm 1. Then we show (lemma 5.3) that the probabilities of vectors $\vec{a}$ and $\vec{b}$ are equal at time t.

In the third step (lemma 5.4) we show that the resequencing buffer occupancy induced by vector $\vec{a}$ is equal to the resequencing buffer occupancy induced by vector $\vec{b}$.

Lemma 5.2: If packet y is transmitted by source S_i, using Algorithm 2, then Algorithm 1 assigns packet $A + y$ to be transmitted by the same source.

Proof: We begin by proving the lemma for $y = 1$ and show that the respective source is S_N.

Algorithm 1: When the first packet is assigned to S_N, we have $x_i \geq x_N \ \ \forall i$, where since this is the first assignment to S_N, $x_N = \overline{d}_N$. Therefore, from the above and the definition of α_i, we conclude that $x_i = \overline{d}_i + \alpha_i$, which means that exactly α_i packets were already assigned to S_i. Consequently, exactly $A = \sum_{i=1}^{N-1} \alpha_i$ packets were assigned to $S_1, S_2, \ldots, S_{N-1}$, when the first packet is assigned to S_N. This proves that the first packet assigned to source S_N is packet $A + 1$.

Algorithm 2: Packet 1 is assigned to the source with the minimal resulting average delay. If we denote by $\tilde{d}_i$ the average delay of packets transmitted by every source $1 \leq i \leq N - 1$, we have $\tilde{d}_i = \overline{d}_i + \alpha_i$ and $\tilde{d}_N = d_N$. From the definition of α_i we learn that $\tilde{d}_N \leq \tilde{d}_i$, $\forall i$. Therefore, packet 1 will be assigned to S_N.

We now have to show similar correspondence for the rest of the sources for packets $2, 3, \ldots, N$.

Algorithm 1: From line 7 of the algorithm, we learn that packet $A + 2$ will be assigned to the source that packet $A + 2 - N$ was assigned to, say S_w. When this assignment took place, x_w was $\overline{d}_w + \alpha_w - 1$, which must have been the lowest of all x_i, at that time. Since this assignment exactly one packet was assigned to every source, therefore, x_i was increased by exactly 1, for all i. Consequently, when packet $A + 2$ is assigned

to S_w, x_w is again the minimal of all x_i. We can easily verify that this property holds when assigning packets $A+3, A+4, \ldots, A+N$ to the sources.

Algorithm 2: Packet 2 is assigned to the source with the minimal average delay among $S_1, S_2, \ldots, S_{N-1}$. From the definition of $\tilde{d}_i$ we notice that the minimum is achieved for S_w, as $\overline{d}_i+\alpha_i$ is minimal for $i=w$, as we proved above. Therefore, packet 2 will be assigned to S_w. Similarly, it can be proved that packets $3, 4, \ldots, N$ will be assigned by Algorithm 2 to the same sources as packets $A+3, A+4, \ldots, A+N$, respectively, are assigned using Algorithm 1.

Finally, Algorithm 2 assigns packets cyclically to sources. Algorithm 1 assigns packets cyclically, after the first A packets are assigned. Therefore, from the above we can conclude that Algorithm 1 assigns packet $A+y$ to the same source as packet y is assigned to by Algorithm 2, for any $y, y \geq 1$. □

Lemma 5.3: The probabilities of vectors $\vec{a}$ and $\vec{b}$ are equal at time t.

proof: We first express the respective probabilities and then show they are equal.

Algorithm 1:

$$Pr_t^1(\vec{b}) = \prod_{i=1}^{N} \sum_{d=d_{min}^i}^{t-\lfloor \frac{a_i-1}{N} \rfloor - \alpha_i} P_i^d \sum_{\delta=t-\lfloor \frac{a_i-1}{N} \rfloor - \alpha_i - d}^{d_{max}^i - d} p_{d,d+\delta}^i \tag{35}$$

where d_{min}^i and d_{max}^i are the minimal and maximal possible delay of a packet transmitted by S_i, respectively. P_i^d and $p_{d,d+\delta}^i$ are the steady state probability that the delay of a packet transmitted by S_i is d and the transition probability of the delay of S_i from d to $d+\delta$.

Expression (35) is obtained as follows:

As explained before, exactly α_i packets were assigned to S_i, till packet $A+1$ was assigned to S_N. Consequently, the first packet assigned to S_i after packet $A+1$ was assigned to S_N is transmitted at time α_i. Packet $A+a_i$ is, therefore, transmitted by S_i at time $\lfloor \frac{a_i-1}{N} \rfloor + \alpha_i$. In order for this packet to arrive till time t, the delay of this packet, d, could be any $d_{min}^i \leq d \leq t - \lfloor \frac{a_i-1}{N} \rfloor - \alpha_i$. The packet transmitted by S_i, right after $A+a_i$, packet $A+a_i+N$, should not arrive till time t. This packet is transmitted at time $\lfloor \frac{a_i-1}{N} \rfloor + \alpha_i + 1$. Given that the delay of packet $A+a_i$ was d, the delay of packet $A+a_i+N$ is $d+\delta$, where $d+\delta+\lfloor \frac{a_i-1}{N} \rfloor + \alpha_i + 1 > t$ must hold. Therefore, the possible values of δ, given d, are $t-d-1-\lfloor \frac{a_i-1}{N} \rfloor - \alpha_i < \delta \leq d_{max}^i - d$, or, $t-d-\lfloor \frac{a_i-1}{N} \rfloor - \alpha_i \leq \delta \leq d_{max}^i - d$.

The probability that the last packet received from S_i, till time t is $A+a_i$, is, therefore, given by

$$\sum_{d=d_{min}^i}^{t-\lfloor \frac{a_i-1}{N} \rfloor - \alpha_i} P_i^d \sum_{\delta=t-\lfloor \frac{a_i-1}{N} \rfloor - \alpha_i - d}^{d_{max}^i - d} p_{d,d+\delta}^i$$

Since the delays of packets transmitted by the different sources are independent, the probability that the

highest indexed packets received from $S_1, S_2, \ldots, S_N$ till time t are $A+a_1, A+a_2, \ldots, A+a_N$, respectively, is simply the product of the corresponding probabilities for all $i, 1 \leq i \leq N$. Therefore, (35) follows.

Algorithm 2:

$$Pr_t^2(\vec{a}) = \prod_{i=1}^{N} \sum_{d=\tilde{d}_{min}^i}^{t-\lfloor \frac{a_i-1}{N} \rfloor} \tilde{P}_i^d \sum_{\delta=t-\lfloor \frac{a_i-1}{N} \rfloor - d}^{\tilde{d}_{max}^i - d} \tilde{p}_{d,d+\delta}^i \tag{36}$$

where $\tilde{d}_{min}^i$ and $\tilde{d}_{max}^i$ are the minimal and maximal possible delay of a packet transmitted by S_i, respectively. $\tilde{P}_i^d$ and $\tilde{p}_{d,d+\delta}^i$ are the steady state probability that the delay of a packet transmitted by S_i is d and the transition probability of the delay of S_i from d to $d+\delta$.

Expression (36) is obtained as follows: Packet a_i was transmitted by S_i at time $\lfloor \frac{a_i-1}{N} \rfloor$. In order for this packet to arrive till time t, the delay of this packet, d, could be any $\tilde{d}_{min}^i \leq d \leq t - \lfloor \frac{a_i-1}{N} \rfloor$. The packet transmitted by S_i, right after a_i, packet $a_i + N$, should not arrive till time t. This packet is transmitted at time $\lfloor \frac{a_i-1}{N} \rfloor + 1$. Given that the delay of packet a_i was d, the delay of packet $a_i + N$ is $d+\delta$, where $d + \delta + \lfloor \frac{a_i-1}{N} \rfloor + 1 > t$ must hold. Therefore, the possible values of δ, given d, are $t - d - 1 - \lfloor \frac{a_i-1}{N} \rfloor < \delta \leq \tilde{d}_{max}^i - d$, or, $t - d - \lfloor \frac{a_i-1}{N} \rfloor \leq \delta \leq \tilde{d}_{max}^i - d$.

The probability that the last packet received from S_i, till time t is a_i, is, therefore, given by

$$\sum_{d=\tilde{d}_{min}^i}^{t-\lfloor \frac{a_i-1}{N} \rfloor} \tilde{P}_i^d \sum_{\delta=t-\lfloor \frac{a_i-1}{N} \rfloor - d}^{\tilde{d}_{max}^i - d} \tilde{p}_{d,d+\delta}^i$$

Since the delays of packets transmitted by the different sources are independent, the probability that the highest indexed packets received from $S_1, S_2, \ldots, S_N$ till time t are $a_1, a_2, \ldots, a_N$, respectively, is simply the product of the corresponding probabilities for all $i, 1 \leq i \leq N$. Therefore, (36) follows.

We now show that $Pr_t^1(\vec{b}) = Pr_t^2(\vec{a})$. From the definition of Algorithm 2 the following equalities emerge instantly:

(1) $d_{min}^i = \tilde{d}_{min}^i - \alpha_i$

(2) $d_{max}^i = \tilde{d}_{max}^i - \alpha_i$

(3) $P_i^d = \tilde{P}_i^{d+\alpha_i}$

(4) $p_{a,b}^i = \tilde{p}_{a+\alpha_i,b+\alpha_i}^i$

Substituting these equalities into (35) we have

$$Pr_t^1(\vec{b}) = \prod_{i=1}^{N} \sum_{d=\tilde{d}_{min}^i-\alpha_i}^{t-\lfloor \frac{a_i-1}{N} \rfloor-\alpha_i} \tilde{P}_i^{d+\alpha_i} \sum_{\delta=t-\lfloor \frac{a_i-1}{N} \rfloor-\alpha_i-d}^{\tilde{d}_{max}^i-\alpha_i-d} \tilde{p}_{d+\alpha_i,d+\alpha_i+\delta}^i =$$

$$\prod_{i=1}^{N} \sum_{d=\tilde{d}^i_{min}}^{t-\lfloor\frac{a_i-1}{N}\rfloor} \tilde{P}_i^d \sum_{\delta=t-\lfloor\frac{a_i-1}{N}\rfloor-d}^{\tilde{d}^i_{max}-d} \tilde{p}^i_{d,d+\delta} = Pr_t^2(\vec{a}) \tag{37}$$

Therefore, the probabilities are equal. □

Lemma 5.4: The resequencing buffer occupancy induced by vector $\vec{a}$ is equal to the resequencing buffer occupancy induced by vector $\vec{b}$.

Proof:

Algorithm 2: Denote $a_m = min_{1\le i\le N}\{a_i\}$, the *mvp* is $a_m + N$ and let the source of the *mvp* be S_m. Every packet indexed higher than the *mvp* that arrived, is stored in the resequencing buffer. The number of packets indexed higher than the *mvp* that were transmitted by S_i and arrived till time t is given by

$$\tilde{c}_i = \lfloor\frac{a_i - a_m}{N}\rfloor$$

Algorithm 1: The source of the *mvp* is, again, S_m, since $min_{1\le i\le N}\{A + a_i\} = A + a_m$, and the *mvp* is $A + a_m + N$. Every packet indexed higher than the *mvp* that arrived is stored in the resequencing buffer. The number of packets indexed higher than the *mvp* that were transmitted by S_i and arrived till time t is given by

$$c_i = \lfloor\frac{(A + a_i) - (A + a_m)}{N}\rfloor = \lfloor\frac{a_i - a_m}{N}\rfloor$$

Since, as indicated above, the contribution of every source to the total buffer occupancy is the same for both algorithms, the lemma follows. □

Proof of theorem 5.1: The theorem follows since we have shown that for every possible arrival scenario, when Algorithm 2 is used, there exists a unique corresponding arrival scenario, when Algorithm 1 is used (lemma 5.2), that occurs with equal probability (lemma 5.3), and results in the same buffer occupancy (lemma 5.4). □

Using Theorem 5.1 we can apply (18) in order to find the resequencing-buffer occupancy probabilities, when packets are assigned to sources according to Algorithm 1, by replacing all the delay probabilities by the shifted-delay probabilities, as explained for Algorithm 2.

VI. Numerical Results

Fig. 11,12 and 13 demonstrate the buffer occupancy probabilities for several scenarios. In all cases, the delay process of each source is represented by a Markov-chain as in Fig. 10, with the transition probabilities and M as parameters of the source.

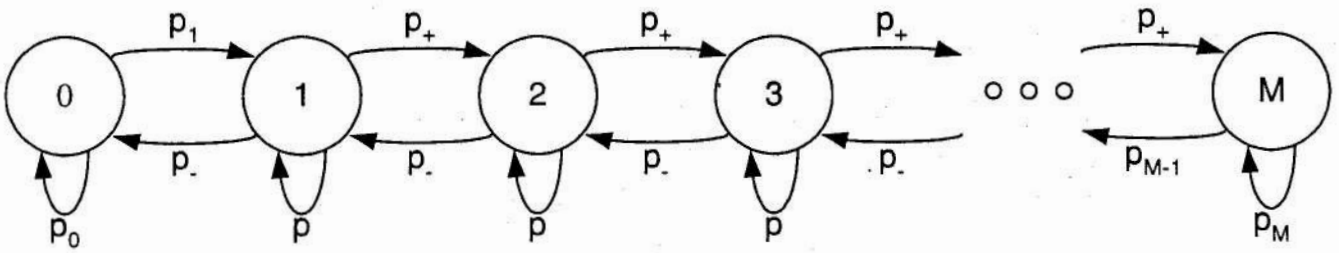


Fig. 10. The simple, finite Markov-chain representing the delay

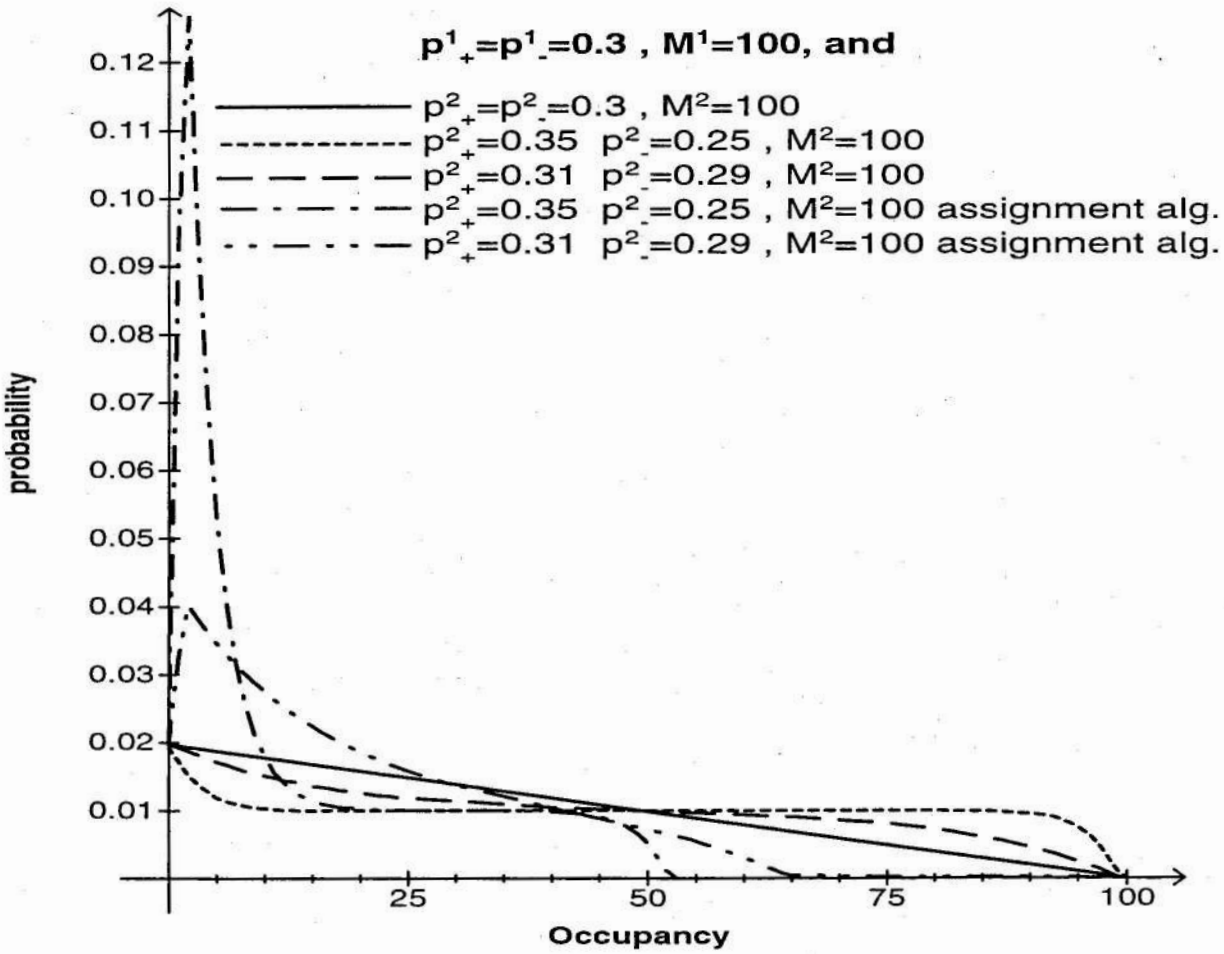


Fig. 11. Buffer occupancy probability - two sources

Fig. 11 demonstrates the buffer occupancy probabilities for several cases of two-source scenarios. The solid line represents the case where the delay models used for both sources is uniform (this is achieved by using $p_+ = p_-$) in the range $[0, 100]$ time units.

For the cases where for source 2, $p_+ = 0.35$, $p_- = 0.25$, we notice how using the assignment algorithm helps in keeping the buffer occupancy low. The average occupancy decreases from 46.3 packets to 14.04 packets when the assignment algorithm is applied.

For the cases where for source 2, $p_+ = 0.31$, $p_- = 0.29$, we, again, notice how using the assignment

algorithm helps in keeping the buffer occupancy low. The average occupancy decreases from 38.7 packets to 20.85 packets when the assignment algorithm is applied.

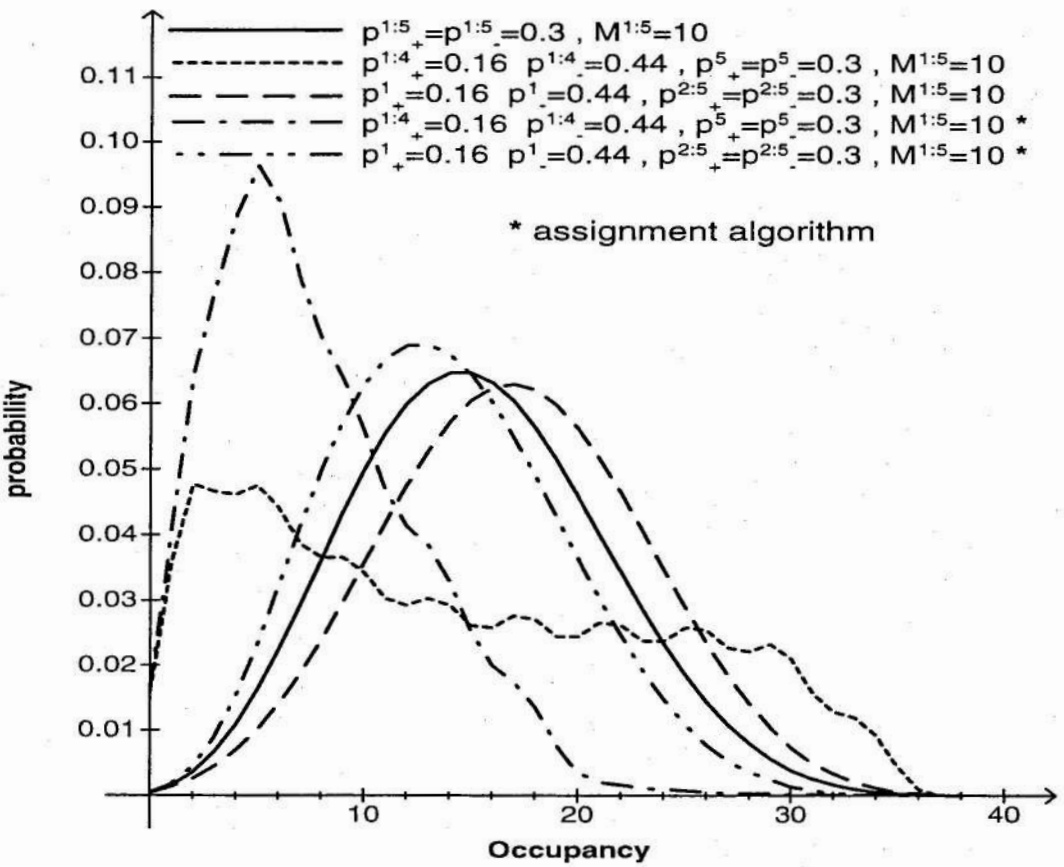


Fig. 12. Buffer occupancy probability - five sources

Fig. 12 demonstrates the buffer occupancy probabilities for several cases of five-source scenarios. The solid line represents the case where the delay models used for all sources is uniform (this is achieved by using $p_+ = p_-$) in the range $[0, 10]$ time units.

For the cases where for sources 1 through 4, $p_+ = 0.16$, $p_- = 0.44$, we can notice how using the assignment algorithm helps in keeping the buffer occupancy low. The average occupancy decreases from 14.27 packets to 7.68 packets when the assignment algorithm is applied.

The cyclic pattern that emerges when the assignment algorithm is not applied is due to the fact that in this scenario, the source of the *mvp* is source 5, with large probability, since the average delay of packets transmitted by this source is 5, while the average delay of packets transmitted by sources 1 through 4 is 1.007. When this occurs, packets stored in the resequencing buffer are packets transmitted by sources 1 through 4. Each source contributes packets to the resequencing-buffer, corresponding to the difference between the delay of the source of the *mvp* and its own delay. Therefore, if, for instance, all four sources have the the same

delay, the resulting occupancy will be a multiple of 4, if three of the four sources have the same delay, and the delay of the fourth is greater by 1, the occupancy will be of the form $4n+1$, $n = 1, 2, 3, \ldots$. Consequently, the occupancy probability has a cyclic pattern, with cycle 4.

However, when the assignment algorithm is applied, the probability of source 5 being the source of the *mvp* is decreased significantly, since, as stated in Lemma 5.2, this case is equivalent to the case where the average delay is the same for all the sources, thus, the cyclic pattern disappears.

For the cases where for source 1, $p_+ = 0.16$, $p_- = 0.44$, we, again, notice how using the assignment algorithm helps in keeping the buffer occupancy low. The average occupancy decreases from 17.04 packets to 13.79 packets when the assignment algorithm is applied. Evidently, the improvement, when using the assignment algorithm, is much less significant than in the former case. This is because the new assignment effects only source 1 in this scenario, while in the former, all the sources 1 through 4 where effected.

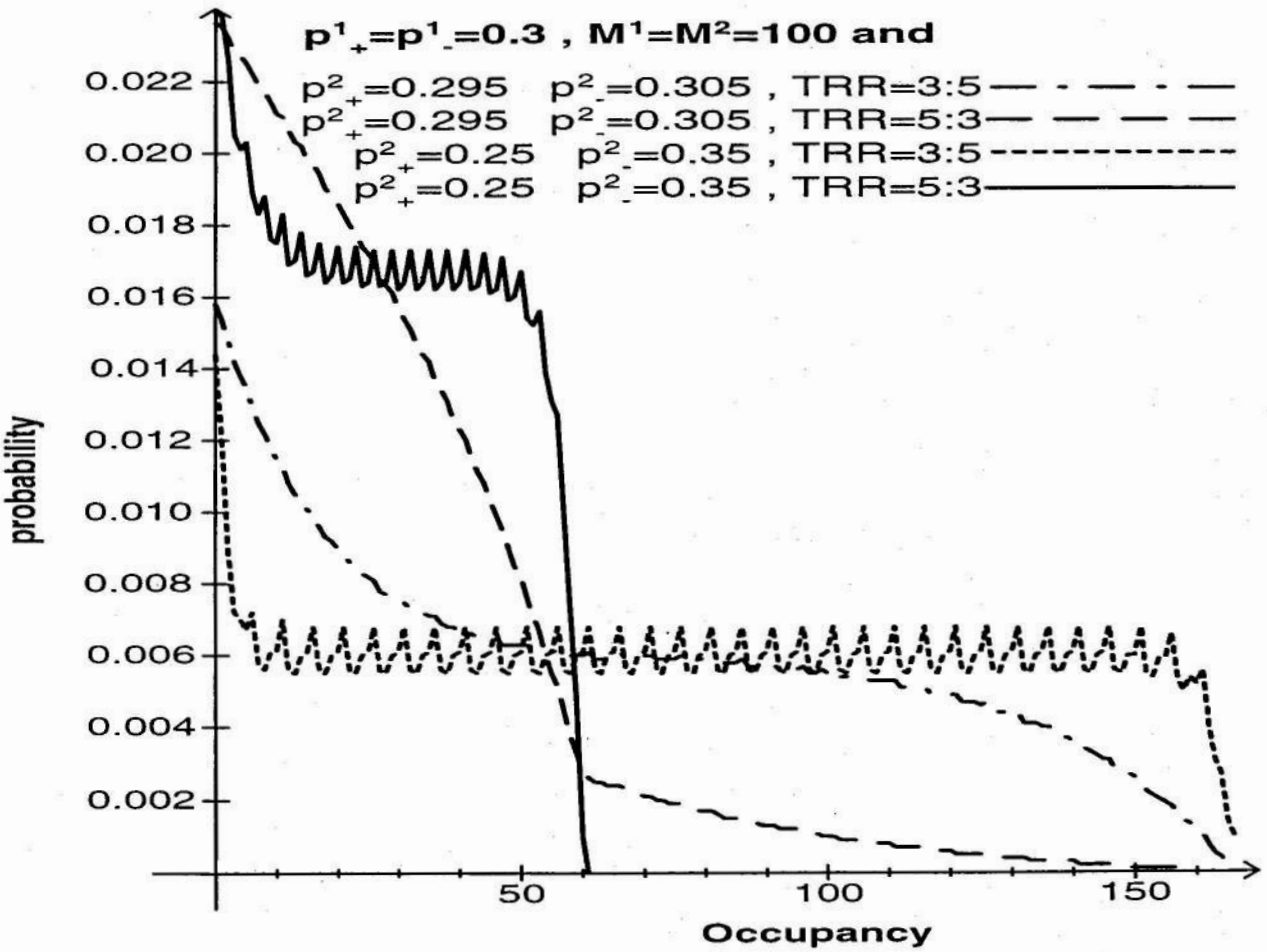


Fig. 13. Buffer occupancy probability - two sources with different transmission rates

Fig. 13 demonstrates the buffer occupancy probabilities for several scenarios where two sources transmit packets, with different packet rates. The TRR is the Transmission Rate Ratio (i.e., if $TRR = 3:5$, source 1

transmits 3 packets in a global time unit, while source 2 transmits 5 packets in a global time unit).

The delay of packets transmitted by source 1 is uniformly distributed in $[0, 100]$ source 1 time units, since $p_+ = p_-$. Therefore, the average delay of packets transmitted by source 1 is 50 source 1 time units.

For the case where for source 2, $p_+ = 0.295$, $p_- = 0.305$, the average delay of packets transmitted by source 2 is 26.2 source 2 time units. Therefore, when the $TRR = 3 : 5$, the average delay of packets transmitted by source 1 is $\frac{50}{3} = 16\frac{2}{3}$ global time units, while the average delay of packets transmitted by source 2 is $\frac{26.2}{5} = 5.24$ global time units. Consequently, the source of the *mvp* is source 1, with very large probability. On the other hand, when $TRR = 5 : 3$, the average delay of packets transmitted by source 1 is $\frac{50}{5} = 10$ global time units, while the average delay of packets transmitted by source 2 is $\frac{26.2}{3} = 8.73$ global time units. Consequently, the source of the *mvp* may be source 1 or source 2, with similar probabilities. Anyway, if the *mvp* was transmitted by source 1, the occupancy would have an upper limit of 60, since the maximal occupancy will be reached if the delay of source 1 was at its maximum, i.e., 100 source 1 time units, and, the delay of source 2 was at its minimum, i.e., 0 source 2 time units. In this case the packets stored in the resequencing buffer are all the packets transmitted by source 2 since the last received packet from source 1 was transmitted. Therefore, given that the delay of source 1 is at its maximum, i.e., $\frac{100}{5} = 20$ global time units, the maximum number of packets stored in the buffer is exactly $20 \cdot 3 = 60$ packets. This explains the dramatic slope change at occupancy 60, since occupancy under 60 may be reached if the source of the *mvp* is either source 1 or 2, while occupancy over 60 may be reached only if the source of the *mvp* packet is source 2.

For the case where for source 2, $p_+ = 0.25$, $p_- = 0.35$, the average delay of packets transmitted by source 2 is 2.9 source 2 time units. Therefore, when $TRR = 3 : 5$, the average delay of packets transmitted by source 1 is $\frac{50}{3} = 16\frac{2}{3}$ global time units, while the average delay of packets transmitted by source 2 is $\frac{2.9}{5} = 0.58$ global time units. Consequently, the source of the *mvp* is source 1, with very large probability.

When $TRR = 5 : 3$, the average delay of packets transmitted by source 1 is $\frac{50}{5} = 10$ global time units, while the average delay of packets transmitted by source 2 is $\frac{2.9}{3} = 0.97$ global time units. Consequently, the source of the *mvp* is, again, source 1, with very large probability. As explained earlier, when $TRR = 5 : 3$, and the source of the *mvp* is source 1, maximal occupancy is 60, which explains the dramatic drop in the occupancy probability near occupancy 60.

In both TRR cases a cyclic pattern is noticed. This pattern is a result of having a very low variance in the delay of packets transmitted by source 2, which is the source contributing to the buffer occupancy. When $TRR = 3 : 5$, 5 packets are transmitted by source 2 every time unit, therefore, the cycle period is 5 packets, while, when $TRR = 5 : 3$, 3 packets are transmitted by source 2 every time unit, therefore, the cycle period is 3 packets.

VII. Summary and Future Work

This paper analyzes the approach of receiving data packets of resources in parallel from several servers, where the resource is available identically, through requesting transmission of specific portions of the resource from each relevant server. The model assumed a constant bit rate from each source, and a stationary delay model, where packets received from each source arrive in the order transmitted.

Closed form equations for calculating the resequencing-buffer occupancy distribution while downloading a resource in parallel from several sources were presented, for homogeneous sources, and for sources with different transmission rates. In order to minimize the resequencing-buffer occupancy, to achieve a more regulated flow of data to the application at the destination, lower the resequencing-buffer demands and to avoid many more unfavorable consequences of out-of-order arrivals, as explained in [11], an algorithm for assigning packets to be transmitted from each source has been devised.

The performance of the algorithm was analyzed. Numerical results show improved performance, especially when the delays of packets transmitted by the different sources are likely to differ significantly.

Interesting questions for future work, regarding the analysis of this approach, include investigating other delay models, including ones that do not assume orderly arrival of packets from a single source, or constant packet rates. Another interesting problem is to model a case where the paths from the sources to the destination are not bottleneck disjoint, which results in dependence in the delays of packets transmitted by the different sources. The algorithm presented here optimizes the buffer occupancy if we assume only static knowledge of the average delay. A more efficient algorithm may take advantage of temporal measurement of the delay, and the delay distribution itself, while taking the accuracy of the measurements into account.

Appendix

Lemma: Given δ_1 and δ_2 at time $t > max\{M_1, M_2+\alpha\}$ (this ensures packets that were assigned in the cyclic assignment part of the packet assignment algorithm to both sources have begun to arrive), the resequencing-buffer occupancy, B, using the packet assignment algorithm is given by:

$$B = \begin{cases} \delta_1 - \delta_2 - \alpha & \delta_1 - \delta_2 \geq \alpha \\ \delta_2 - \delta_1 - 1 + \alpha & \delta_1 - \delta_2 < \alpha \end{cases} \tag{38}$$

Proof: At time $t > max(M_1, M_2 + \alpha)$ source 1 transmits packet $2t + \alpha + 1$, and source 2 transmits packet $\alpha + 2 + 2(t - \alpha) = 2t - \alpha + 2$. Given δ_1 and δ_2, the last packet received from source 1 is packet $L_1 - 2t + \alpha + 1 - 2\delta_1$, and the last packet received from source 2 is packet $L_2 = 2t - \alpha + 2 - 2\delta_2$.

If $\delta_1 - \delta_2 \geq \alpha$ then $L_2 > L_1$, therefore, if there are any packets in the resequencing-buffer they are packets transmitted by source 2, and the resequencing-buffer occupancy is

$$B = \frac{L_2 - (L_1 + 1)}{2} = \frac{2t - \alpha + 2 - 2\delta_2 - 2t - \alpha - 1 + 2\delta_1 - 1}{2} = \delta_1 - \delta_2 - \alpha \qquad (39)$$

since packet L_1+1 (transmitted by source 2) is not stored in the resequencing-buffer, because all its preceding packets have arrived. All the packets that were transmitted by source 2 after packet $L_1 + 1$, and arrived till time t are stored in the resequencing-buffer. Since no other packets transmitted by source 2 and no packet transmitted by source 1 are stored in the resequencing-buffer, (39) holds.

If $\delta_1 - \delta_2 < \alpha$ then $L_1 > L_2$, therefore, if there are any packets in the resequencing-buffer they are packets transmitted by source 1, and the resequencing-buffer occupancy is

$$B = \frac{L_1 - (L_2 + 1)}{2} = \frac{2t + \alpha + 1 - 2\delta_1 - 2t + \alpha - 2 + 2\delta_2 - 1}{2} = \delta_2 - \delta_1 - 1 + \alpha \qquad (40)$$

since packet L_2+1 (transmitted by source 1) is not stored in the resequencing-buffer, because all its preceding packets have arrived. All the packets that were transmitted by source 1 after packet $L_2 + 1$, and arrived till time t are stored in the resequencing-buffer. Since no other packets transmitted by source 1 and no packet transmitted by source 2 are stored in the resequencing-buffer, (40) holds. Therefore, (38) holds, and the proof is completed. □

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